

Lecture Notes

EC226 Term 1 2025/26

Structural change (Chow Tests 1 and 2)

Readings:

- Stock and Watson (2003): Chapter 14
- Dougherty (2016): Chapter 5
- Wooldridge (2013): Chapter 7

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1 Properties of estimators

Consider the model:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i \quad i = 1, 2, \dots, n \quad (1)$$

where the CLRM assumptions hold. As the model holds for all n observations:

$$\frac{\partial y_i}{\partial x_{ji}} = \beta_j \quad \forall i = 1, \dots, n \quad (\text{holding all else constant})$$

If the coefficients of the equation (1) are not constant over the entire sample, but for the first set on n_1 observations we have:

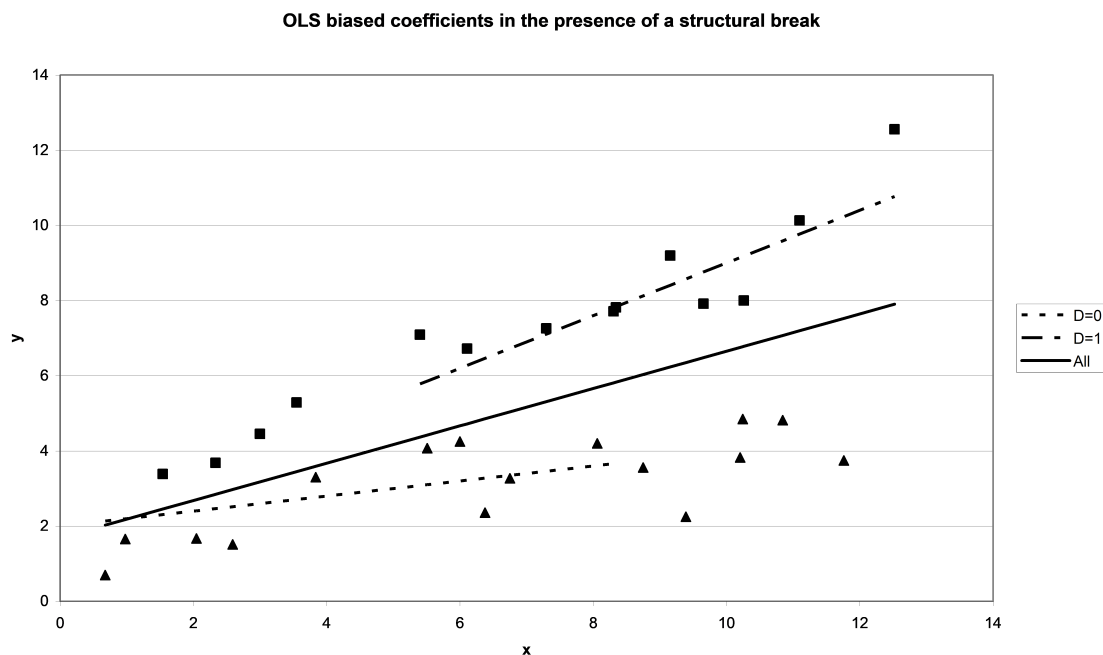
$$y_i = \beta_0^1 + \beta_1^1 x_{1i} + \beta_2^1 x_{2i} + \dots + \beta_k^1 x_{ki} + \varepsilon_i^1 \quad i = 1, 2, \dots, n_1 \quad (2)$$

and for the second set of $n - n_1$ observations we have:

$$y_i = \beta_0^2 + \beta_1^2 x_{1i} + \beta_2^2 x_{2i} + \dots + \beta_k^2 x_{ki} + \varepsilon_i^2 \quad i = n_1 + 1, 2, \dots, n \quad (3)$$

In which case estimating equation (1) will yield biased estimates of the true coefficients, for example, the OLS estimator $\hat{\beta}_j$ from equation (1) will be some weighted average of the unbiased estimators $\hat{\beta}_j^1$ and $\hat{\beta}_j^2$.

We can see this diagrammatically in the Figure below, in this figure the solid line is the OLS coefficient estimates over the entire sample of 30 observations (with a slope estimate of 0.5). However, the parameters are not constant over the entire sample the triangular points come from one sample (with slope coefficient 0.2 – estimated with dotted line) and the square points come from a separate sample (with slope coefficient 0.7 – estimated with dashed-dotted line). Clearly the estimate of 0.5 is a biased estimate of the slope coefficient both samples.



2 Detecting structural change

2.1 Structural Change test (1st Chow Test)

To test the null hypothesis that equation (1) is correct and the parameters are constant over the entire sample of observations, then the parameters in equations (2) and (3) are equal:

$$H_0 : \beta_0^1 = \beta_0^2, \beta_1^1 = \beta_1^2, \beta_2^1 = \beta_2^2, \dots, \beta_k^1 = \beta_k^2 \quad (4)$$

Imposing this null hypothesis on equations (2) and (3) yields equation (1). Equation (1) is therefore the restricted equation and equations (2) and (3) (taken together) are the unrestricted equations. The test for constant parameters (or no structural break), corresponding to H_0 above, is an F-test:

$$F = \frac{[RSS^R - RSS^U]/d}{RSS^U/DoF} = \text{Chow Test} \quad (5)$$

where $RSS^R = RSS^{(1)}$ (where the superscript (1) indicates the RSS is from equation (1), $RSS^U = RSS^{(2)} + RSS^{(3)}$, $d = (k+1)$ (number of equality signs in H_0 (or number of parameters in unrestricted model $2(k+1)$) minus number of parameters in restricted model $(k+1)$) and $DoF = n - 2(k+1)$ (degrees of freedom in unrestricted model).

An alternative way of undertaking the same test would be through the use of dummy variables. Consider the equation:

$$y_i = \delta_0 + \delta_1 x_{1i} + \delta_2 x_{2i} + \dots + \delta_k x_{ki} + \gamma_0 D_i + \gamma_1 x_{1i} D_i + \gamma_2 x_{2i} D_i + \dots + \gamma_k x_{ki} D_i + \zeta_i \quad (6)$$

where $D_i = 1$ if $i = n_1 + 1, \dots, n$ and 0 otherwise, is an additive dummy variable and $D_i x_{ji}$ $j = 1, \dots, k$ are a series of multiplicative dummy variables. To understand dummy variables it is best to write out equation (6) for all categories of the dummy variable:

Regime 1: $i = 1, 2, \dots, n_1$ ($D_i = 0$)

$$y_i = \delta_0 + \delta_1 x_{1i} + \delta_2 x_{2i} + \dots + \delta_k x_{ki} + \zeta_i$$

this equation is identical to that in equation (2), therefore, $\delta_0 = \beta_0^1, \delta_1 = \beta_1^1, \delta_2 = \beta_2^1, \dots, \delta_k = \beta_k^1$.

Regime 2: $i = n_1 + 1, \dots, n$ ($D_i = 1$)

$$y_i = \delta_0 + \delta_1 x_{1i} + \delta_2 x_{2i} + \dots + \delta_k x_{ki} + \gamma_0 + \gamma_1 x_{1i} + \gamma_2 x_{2i} + \dots + \gamma_k x_{ki} + \zeta_i$$

And collecting similar terms together

$$y_i = (\delta_0 + \gamma_0) + (\delta_1 + \gamma_1)x_{1i} + (\delta_2 + \gamma_2)x_{2i} + \dots + (\delta_k + \gamma_k)x_{ki} + \zeta_i$$

This equation is identical to that in equation (3) and therefore, $\delta_0 + \gamma_0 = \beta_0^2, \delta_1 + \gamma_1 = \beta_1^2, \delta_2 + \gamma_2 = \beta_2^2, \dots, \delta_k + \gamma_k = \beta_k^2$.

Rearranging these equations and using the results from Regime 1 we have:

$$\gamma_0 = \beta_0^2 - \delta_0 \Rightarrow \gamma_0 = \beta_0^2 - \beta_0^1 = \text{Intercept shift}$$

$$\gamma_1 = \beta_1^2 - \delta_1 \Rightarrow \gamma_1 = \beta_1^2 - \beta_1^1 = \text{Coefficient shift on } x_1$$

$$\gamma_2 = \beta_2^2 - \delta_2 \Rightarrow \gamma_2 = \beta_2^2 - \beta_2^1 = \text{Coefficient shift on } x_2$$

$$\gamma_k = \beta_k^2 - \delta_k \Rightarrow \gamma_k = \beta_k^2 - \beta_k^1 = \text{Coefficient shift on } x_k$$

In which case, the null hypothesis written in equation (4) is $H_0 : \gamma_0 = \gamma_1 = \gamma_2 = \dots = \gamma_k = 0$. Imposing this null hypothesis on equation (6), we get

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i \quad i = 1, 2, \dots, n$$

which is exactly equation (1). The F-test for testing $H_0 : \gamma_0 = \gamma_1 = \gamma_2 = \dots = \gamma_k = 0$ is calculated as:

$$F = \frac{(RSS^R - RSS^U)/d}{RSS^U/DoF}$$

where $RSS^R = RSS^{(1)}$, $RSS^U = RSS^{(6)}$, $d = k+1$, and $DoF = n-2(k+1)$. Given that equation (6) has the same parameter estimates as equations (2) and (3) together, then $RSS^{(6)} = RSS^{(2)} + RSS^{(3)}$, and the F-test above is exactly the same as equation (5).

Using the dummy variable approach it is also possible to test that only a subset of the coefficients have changed in the sub-samples. For example, to test that only the intercept and the coefficient on x_{1i} have changed, estimate following equation:

$$y_i = \delta_0 + \delta_1 x_{1i} + \delta_2 x_{2i} + \dots + \delta_k x_{ki} + \gamma_0 D_i + \gamma_1 x_{1i} D_i + \zeta_i \quad (7)$$

and test $H_0 : \gamma_0 = \gamma_1 = 0$ via the F-test

$$F = \frac{(RSS^R - RSS^U)/d}{RSS^U/DoF}$$

where $RSS^R = RSS^{(1)}$, $RSS^U = RSS^{(7)}$, $d = 2$, $DoF = n - (k+3)$.

NOTE:

See Appendix A for a worked empirical example for testing for structural change and Appendix B for another example question.

2.2 Predictive Failure Test (2nd Chow test)

This tests the hypothesis that an equation estimated over the first n observations holds for the last n_1 observations. The equation estimated over the first n observations is

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i \quad i = 1, 2, \dots, n \quad (8)$$

and the null hypothesis is:

$$H_0 : y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i \quad i = n+1, \dots, n+n_1$$

in which case imposing the null hypothesis on the unrestricted equation (8), yields the restricted model:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i \quad i = 1, 2, \dots, n+n_1 \quad (9)$$

the F-test for Predictive Failure is

$$F = \frac{(RSS^R - RSS^U)/d}{RSS^U/DoF} \quad (10)$$

where $RSS^R = RSS^{(9)}$, $RSS^U = RSS^{(8)}$, $d = n_1$ (number of equality signs in H_0), $DoF = n - (k+1)$. This test can alternatively be undertaken using dummy variables. Consider the equation:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \sum_{j=1}^{n_1} \gamma_j D_{ji} + \varepsilon_i \quad i = 1, 2, \dots, n + n_1 \quad (11)$$

where $D_{ji} = 1$ if $i = n + j$ where $j = 1, \dots, n_1$ and 0 otherwise, are impulse dummy variables taking the value 1 for one particular observation zero for all others. OLS estimation of equation (11) yields $\hat{\gamma}_j$ = Prediction error in period $n + j$ (for a demonstration of this see below).

The null hypothesis is, $H_0 : \gamma_1 = \gamma_2 = \dots = \gamma_{n_1} = 0$ imposing the H_0 on the unrestricted model (11), yields the restricted model:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i \quad i = 1, 2, \dots, n + n_1$$

which is equation (8). The F-test is:

$$F = \frac{(RSS^R - RSS^U)/d}{RSS^U/DoF} \quad (12)$$

where, $RSS^R = RSS^{(9)}$, $RSS^U = RSS^{(11)} (\equiv RSS^{(8)})$, $d = n_1$, $DoF = n - (k + 1)$ and equation (12) gives exactly the same F-statistic as equation (10).

To interpret the coefficient on the dummy variables (D_j), consider the simplest case in which we are only extending the equation by 1 observation (i.e. the new observation is $(n + 1)$ and we only have 1 explanatory variable, x :

$$y_i = \beta_0 + \beta_1 x_i + \gamma_1 D_i + \varepsilon_i \quad i = 1, 2, \dots, n, n + 1$$

where, $D_i = 1$ if $i = n + 1$ and 0 otherwise.

Now we wish to minimise

$$RSS_{n+1} = \sum_{i=1}^{n+1} (y_i - b_0 - b_1 x_i - \hat{\gamma}_1 D_i)^2 = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2 + (y_{n+1} - b_0 - b_1 x_{n+1} - \hat{\gamma}_1)^2 \quad (13)$$

with respect to the three parameters: b_0 , b_1 , $\hat{\gamma}_1$. Differentiating with respect to the parameter $\hat{\gamma}_1$ and setting the expression to zero:

$$\begin{aligned} \frac{\partial \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2 + (y_{n+1} - b_0 - b_1 x_{n+1} - \hat{\gamma}_1)^2}{\partial \hat{\gamma}_1} &= \frac{\partial (y_{n+1} - b_0 - b_1 x_{n+1} - \hat{\gamma}_1)^2}{\partial \hat{\gamma}_1} \\ &= -2(y_{n+1} - b_0 - b_1 x_{n+1} - \hat{\gamma}_1) = 0 \\ &\Rightarrow \hat{\gamma}_1 = y_{n+1} - (b_0 + b_1 x_{n+1}) \end{aligned} \quad (14)$$

where e_{n+1} is the residual in period $(n + 1)$, y_{n+1} is the actual value of y in period $(n + 1)$, $\hat{y}_{n+1} = b_0 + b_1 x_{n+1}$ is the OLS predicted value for y in period $(n + 1)$ and therefore $\hat{\gamma}_1$ = Prediction error.

NOTE:

The second term in RSS_{n+1} in equation (13) is zero (given equation (14)), in which case $RSS_{n+1} = RSS_n$. The derivative of the RSS_{n+1} with respect to b_0 yields

$$\frac{\partial \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2 + \overbrace{(y_{n+1} - b_0 - b_1 x_{n+1} - \hat{\gamma}_1)^2}^{e_{n+1}=0}}{\partial b_0} = \frac{\partial \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2}{\partial b_0} \quad (15)$$

which is the simple two variable regression problem. Similarly the derivative of the RSS_{n+1} with respect to b_1 yields

$$\frac{\partial \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2 + \overbrace{(y_{n+1} - b_0 - b_1 x_{n+1} - \hat{\gamma}_1)^2}^{e_{n+1}=0}}{\partial b_1} = \frac{\partial \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2}{\partial b_1} \quad (16)$$

which is again the simple two variable regression problem. Consequently, the solutions to equations (15) and (16) must be the solutions to NORMAL equations in the two variable regression model:

$$b_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \text{ and } b_0 = \bar{y} - b_1 \bar{x}$$

NOTE

In general, given the equation:

$$y_i = \beta_0 + \beta_1 x_i + \sum_{j=1}^{n_1} \gamma_j D_{ji} + \varepsilon_i \quad i = 1, 2, \dots, n, n+1, \dots, n+n_1$$

where $D_{ji} = 1$ when $i = n+j$ for $j = 1, 2, \dots, n_1$ and 0 otherwise, then $\hat{\gamma}_j$ = Prediction error in period $n+j$ and $e_{n+j} = 0$ for $j = 1, \dots, n_1$. See Appendix C for an example of a question related to predictive failure and Chow's 2nd test.

2.3 Outlier Data Points

Suppose you have an equation of the form:

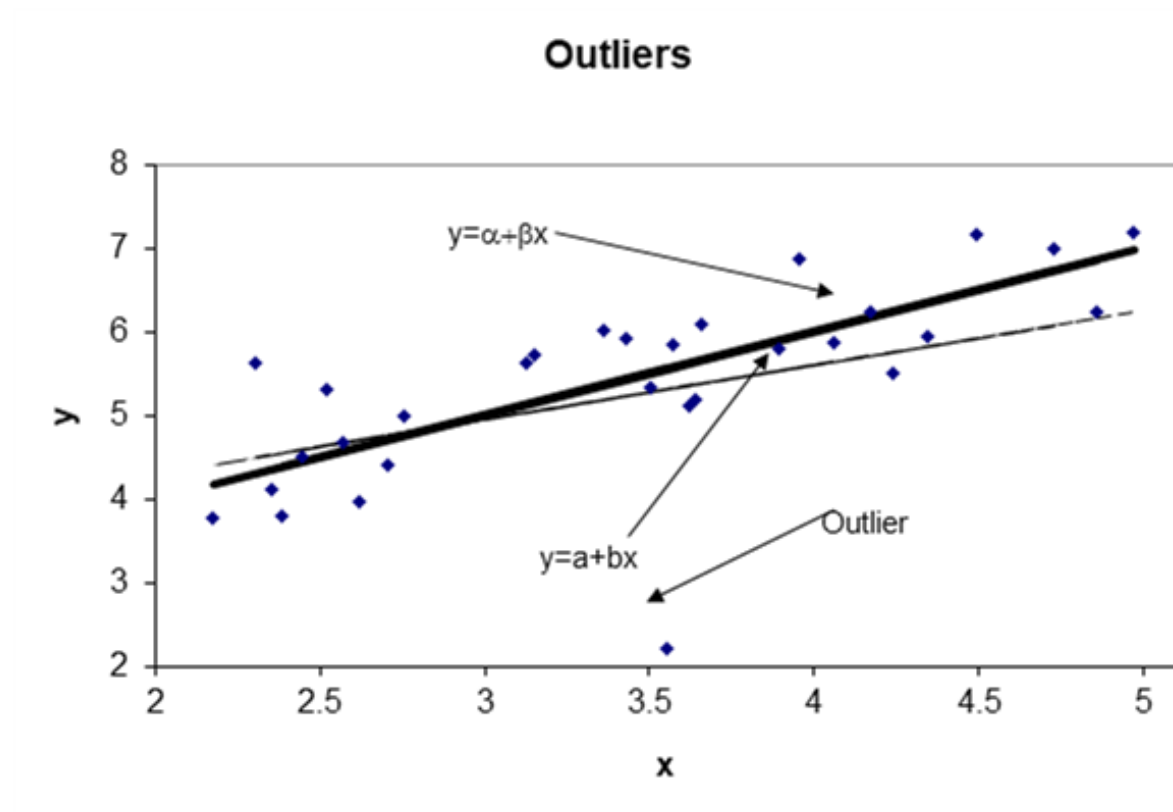
$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad i = 1, \dots, n$$

and there is an outlier in one of the y values due to miscoding, then you end up with a scatter plot as below. The true regression line is indicated by the dark line, whereas the estimated relationship $\hat{y}_i = a + bx$ is biased (the intercept is upward biased and the slope downward biased) due to the influence of the outlying data point – represented by the dashed line.

In this case the solution is to include an impulse dummy variable to the outlier point and this will give you an unbiased estimate of the true regression line. Including an impulse dummy variable (D_i) for the outlier point and estimating:

$$y_i = \beta_0 + \beta_1 x_i + \gamma D_i + \varepsilon_i \quad i = 1, \dots, n$$

is equivalent (exactly the same as) dropping the outlier data point (for both the dependent variable and explanatory variables) from the dataset.



Appendices

A Example of Structural Change

1. ALL Individuals

$$lhourpay_i = \beta_0 + \beta_1 age_i + \beta_2 age_i^2 + \beta_3 edage_i + \varepsilon_i, \quad i = 1, \dots, 9531 \quad (A1)$$

Figure 1: Equation: lhourpay on age age² and edage for all individuals

Source	SS	df	MS	Number of obs	=	9,531
Model	417.196605	3	139.065535	F(3, 9527)	=	518.21
Residual	2556.65724	9,527	.26835911	Prob > F	=	0.0000
				R-squared	=	0.1403
				Adj R-squared	=	0.1400
Total	2973.85385	9,530	.31205182	Root MSE	=	.51803

lhourpay	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
age	.0604697	.0027315	22.14	0.000	.0551155	.065824
c.age#c.age	-.0006095	.0000311	-19.60	0.000	-.0006704	-.0005485
edage	.0510923	.0017506	29.19	0.000	.0476609	.0545238
_cons	.2140556	.0636803	3.36	0.001	.0892288	.3388825

2. MALES

$$lhourpay_i = \beta_0^M + \beta_1^M age_i + \beta_2^M age_i^2 + \beta_3^M edage_i + \varepsilon_i, \quad i = 1, \dots, 4518 \quad (A2)$$

Figure 2: Equation: lhourpay on age age² and edage for males

Source	SS	df	MS	Number of obs	=	4,518
Model	244.285206	3	81.4284022	F(3, 4514)	=	298.05
Residual	1233.2618	4,514	.273208198	Prob > F	=	0.0000
				R-squared	=	0.1653
				Adj R-squared	=	0.1648
Total	1477.54701	4,517	.327108039	Root MSE	=	.52269

lhourpay	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
age	.0708599	.0039074	18.13	0.000	.0631995	.0785203
c.age#c.age	-.0007182	.0000443	-16.22	0.000	-.000805	-.0006314
edage	.0532904	.0025852	20.61	0.000	.0482222	.0583586
_cons	.0550913	.0913374	0.60	0.546	-.1239746	.2341573

3. FEMALES

$$lhourpay_i = \beta_0^F + \beta_1^F age_i + \beta_2^F age_i^2 + \beta_3^F edage_i + \varepsilon_i, \quad i = 1, \dots, 5013 \quad (A3)$$

Figure 3: Equation lhourpay on age age² and edage for females

Source	SS	df	MS	Number of obs	=	5,013
Model	188.221515	3	62.740505	F(3, 5009)	=	258.60
Residual	1215.24872	5,009	.24261304	Prob > F	=	0.0000
				R-squared	=	0.1341
				Adj R-squared	=	0.1336
Total	1403.47023	5,012	.280021993	Root MSE	=	.49256

lhourpay	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
age	.0527563	.0036697	14.38	0.000	.0455621	.0599506
c.age#c.age	-.0005318	.000042	-12.67	0.000	-.000614	-.0004495
edage	.0501494	.0022852	21.94	0.000	.0456693	.0546295
_cons	.3086063	.0852637	3.62	0.000	.1414521	.4757606

4. ALL individuals

$$\begin{aligned}
lhourpay_i = & \delta_0 + \delta_1 age_i + \delta_2 age_i^2 + \delta_3 edage_i + \gamma_0 sex_i + \gamma_1 sex_i \times age_i \\
& + \gamma_2 sex_i \times age_i^2 + \gamma_3 sex_i \times edage_i + \varepsilon_i, \quad i = 1, \dots, 9513
\end{aligned}
\tag{A4}$$

Figure 4: Equation lhourpay on age age² edage sex sex*age sex*age² and sex*edage for all individuals

Source	SS	df	MS	Number of obs	=	9,531
Model	525.343326	7	75.0490466	F(7, 9523)	=	291.89
Residual	2448.51052	9,523	.257115459	Prob > F	=	0.0000
				R-squared	=	0.1767
				Adj R-squared	=	0.1760
Total	2973.85385	9,530	.31205182	Root MSE	=	.50707

lhourpay	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
age	.0708599	.0037906	18.69	0.000	.0634296	.0782903
c.age#c.age	-.0007182	.000043	-16.72	0.000	-.0008024	-.000634
edage	.0532904	.0025079	21.25	0.000	.0483745	.0582064
sex						
Female	.253515	.124722	2.03	0.042	.0090332	.4979968
sex#c.age						
Female	-.0181036	.0053517	-3.38	0.001	-.028594	-.0076132
sex#c.age#c.age						
Female	.0001864	.0000609	3.06	0.002	.000067	.0003059
sex#c.edage						
Female	-.003141	.0034386	-0.91	0.361	-.0098814	.0035994
_cons	.0550913	.0886065	0.62	0.534	-.1185963	.228779

For MALES (from equation (A4)):

$$lhourpay_i = \delta_0 + \delta_1 age_i + \delta_2 age_i^2 + \delta_3 edage_i + u_i$$

Variable	Coefficient
cons	.0550913
age	.0700599
agesq	-.0007182
edage	.0532904

and $\delta_0 = \beta_0^M$, $\delta_1 = \beta_1^M$, $\delta_2 = \beta_2^M$, $\delta_3 = \beta_3^M$ (which exactly match up with the estimates reported in Figure 2. MALES).

For FEMALES (from equation (A4)):

$$lhourpay_i = (\delta_0 + \gamma_0) + (\delta_1 + \gamma_1)age_i + (\delta_2 + \gamma_2)age_i^2 + (\delta_3 + \gamma_3)edage_i + \varepsilon_i$$

Variable	Coef($\hat{\delta}$)	Coef($\hat{\gamma}$)	Coef($\hat{\delta} + \hat{\gamma}$)
<i>cons</i>	.050913	0.253515	0.3086063
<i>age</i>	0.0700599	-0.0181036	0.0527563
<i>age</i> $\times$ <i>age</i>	-0.0007182	0.0001864	-0.0005318
<i>edage</i>	0.0532904	-0.003141	0.0501494

and $\delta_0 + \gamma_0 = \beta_0^F$, $\delta_1 + \gamma_1 = \beta_1^F$, $\delta_2 + \gamma_2 = \beta_2^F$, $\delta_3 + \gamma_3 = \beta_3^F$ (which exactly match up with the results in Figure 3: Females)

NOTE 1:

RSS(M)+RSS(F)=1233.2618 + 1215.24872 = 2448.51052 i.e. RSS Figure 4. All individuals.

NOTE 2:

Equation (A2) + Equation (A3) = Equation (A4).

5. ALL individuals

$$lhourpay_i = \delta_0 + \delta_1 age_i + \delta_2 age_i^2 + \delta_3 edage_i + \gamma_0 sex_i + \varepsilon_i, \quad i = 1, \dots, 9351 \quad (A5)$$

Figure 5: Equation lhourpay on age age² edage sex for all individuals

Source	SS	df	MS	Number of obs	=	9,531
Model	521.362057	4	130.340514	F(4, 9526)	=	506.27
Residual	2452.49179	9,526	.257452424	Prob > F	=	0.0000
				R-squared	=	0.1753
				Adj R-squared	=	0.1750
Total	2973.85385	9,530	.31205182	Root MSE	=	.5074

lhourpay	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
age	.0616758	.0026761	23.05	0.000	.0564301	.0669214
c.age#c.age	-.0006238	.0000305	-20.47	0.000	-.0006835	-.0005641
edage	.0517608	.0017149	30.18	0.000	.0483992	.0551225
sex						
Female	-.2094897	.0104148	-20.11	0.000	-.2299048	-.1890745
_cons	.2887707	.0624833	4.62	0.000	.1662902	.4112513

In summary we have 5 equations above

(a) Equation (A1):

$$lhourpay_i = \beta_0 + \beta_1 age_i + \beta_2 age_i^2 + \beta_3 edage_i + \varepsilon_i, \quad i = 1, \dots, 9531$$

No of parameters= 4, $RSS = 2556.65724$, $DoF = 9531 - 4 = 9527$

(b) Equation (A2):

$$lhourpay_i = \beta_0^M + \beta_1^M age_i + \beta_2^M age_i^2 + \beta_3^M edage_i + \varepsilon_i, \quad i = 1, \dots, 4518$$

No of parameters= 4, $RSS = 1233.26180$, $DoF = 4518 - 4 = 4514$

(c) Equation (A3):

$$lhourpay_i = \beta_0^F + \beta_1^F age_i + \beta_2^F age_i^2 + \beta_3^F edage_i + \varepsilon_i, \quad i = 1, \dots, 5013$$

No of parameters= 4, $RSS = 1215.24872$, $DoF = 5013 - 4 = 5009$

(d) Equation (A4):

$$lhourpay_i = \delta_0 + \delta_1 age_i + \delta_2 age_i^2 + \delta_3 edage_i + \gamma_0 sex_i + \gamma_1 sex_i \times age_i + \gamma_2 sex_i \times age_i^2 \\ + \gamma_3 sex_i \times edage_i + \varepsilon_i, \quad i = 1, \dots, 9531$$

No of parameters= 8, $RSS = 2448.51052$, $DoF = 9531 - 8 = 9523$

(e) Equation (A5):

$$lhourpay_i = \delta_0 + \delta_1 age_i + \delta_2 age_i^2 + \delta_3 edage_i + \gamma_0 sex_i + \varepsilon_i, \quad i = 1, \dots, 9531$$

No of parameters= 5, $RSS = 2452.49179$, $DoF = 9531 - 5 = 9526$

To test for structural change:

Test 1a (comparing equation (A2) + equation (A3) vs equation (A1)):

$$H_0 : \beta_0^M = \beta_0^F, \beta_1^M = \beta_1^F, \beta_2^M = \beta_2^F, \beta_3^M = \beta_3^F$$

$$H_1 : \beta_0^M \neq \beta_0^F, \beta_1^M \neq \beta_1^F, \beta_2^M \neq \beta_2^F, \beta_3^M \neq \beta_3^F$$

$$F = \frac{(RSS^R - RSS^U)/d}{RSS^U/DoF}$$

Restricted equation: equation (A1), $RSS^R = 2556.65724$

Unrestricted equation (equation (A2) + equation (A3)), $RSS^U = 1233.26180 + 1215.24872 = 2448.51052$

$DoF = 9531 - 8$, $d = \text{Number of parameters from unrestricted} - \text{Number of parameters from restricted} = 8 - 4$ (=number of equality signs in H_0)

$$F = \frac{(2556.65724 - 2448.51052)/4}{2448.51052/(9531 - 8)} = 105.15 \quad F_{4,9523}^{0.01} = 3.32 \quad \text{Reject } H_0$$

Test 1b (comparing equation (A4) vs equation (A1)):

$$H_0 : \gamma_0 = \gamma_1 = \gamma_2 = \gamma_3 = 0$$

$$H_1 : \gamma_j \neq 0, j = 1, 2, 3, 4$$

$$F = \frac{(RSS^R - RSS^U)/d}{RSS^U/DoF}$$

Restricted equation: equation (A1), $RSS^R = 2452.49179$

Unrestricted equation: equation (A4), $RSS^U = 2448.51052$ $DoF = 9531 - 8$,

$d = \text{Numbers of parameters from unrestricted} - \text{Number of parameters from restricted} = 8 - 4$ (=number of equality signs in H_0)

$$F = \frac{(2556.65724 - 2448.51052)/4}{2448.51052/(9531 - 8)} = 105.15 \quad F_{4,9523}^{0.01} = 3.32 \quad \text{Reject } H_0$$

and Test 1a and Test 1b are identical as equation (A2) + equation (A3) is the same as equation (A4).

Test 2a (Comparing equation (A2) + equation (A3) vs equation (A5))

$$H_0 : \beta_1^M = \beta_1^F, \beta_2^M = \beta_2^F, \beta_3^M = \beta_3^F$$

$$H_1 : \beta_1^M \neq \beta_1^F, \beta_2^M \neq \beta_2^F, \beta_3^M \neq \beta_3^F$$

$$F = \frac{(2452.49179 - 2448.51052)/3}{2448.51052/(9531 - 8)} = 5.16 \quad F_{3,9523}^{0.01} = 3.79 \quad \text{Reject } H_0$$

Test 2b (Comparing equation (A4) vs equation (A5)):

$$H_0 : \gamma_1 = \gamma_2 = \gamma_3 = 0$$

$$H_1 : \gamma_j \neq 0, j = 1, 2, 3$$

$$F = \frac{(2452.49179 - 2448.51052)/3}{2448.51052/(9531 - 8)} \quad F_{3,9523}^{0.01} = 3.79 \quad \text{Reject } H_0$$

and Test 2a and Test 2b are identical as equation (A2) + equation (A3) is the same as equation (A4).

Stata could have tested these restrictions directly, if after the command estimating the equation described in Figure 4 we had put:

```
. test 2.sex 2.sex#c.age 2.sex#c.age#c.age 2.sex#c.edage

( 1) 2.sex = 0
( 2) 2.sex#c.age = 0
( 3) 2.sex#c.age#c.age = 0
( 4) 2.sex#c.edage = 0

F( 4, 9523) = 105.15
Prob > F = 0.0000

. test 2.sex#c.age 2.sex#c.age#c.age 2.sex#c.edage

( 1) 2.sex#c.age = 0
( 2) 2.sex#c.age#c.age = 0
( 3) 2.sex#c.edage = 0

F( 3, 9523) = 5.16
Prob > F = 0.0015
```

B Exercise of Structural Change

An equation to determine happiness from a random sample of 600 employed individuals is estimated based on the following model

$$\ln(Happy_i) = \alpha + \beta_1 \ln(Income_i) + \beta_2 Female_i + \beta_3 Married_i + \beta_4 SOCI_i + \beta_5 SOCII_i + \beta_6 SOCIII_i + \beta_7 SOCIV_i + \beta_8 Educ_i + \beta_9 Age_i + \varepsilon_i \quad (B1)$$

where: *Happy* = Happiness score (out of 100); *Income*=Annual income (£000s); *Educ* = Number of years in education; *Age* = Age. $Female_i = \begin{cases} 1 & \text{female} \\ 0 & \text{male} \end{cases}$, $Married_i = \begin{cases} 1 & \text{married} \\ 0 & \text{otherwise} \end{cases}$,

$$SOCI_i = \begin{cases} 1 & \text{Professional} \\ 0 & \text{otherwise} \end{cases}, SOCII_i = \begin{cases} 1 & \text{managerial} \\ 0 & \text{otherwise} \end{cases}, SOCIII_i = \begin{cases} 1 & \text{skilled} \\ 0 & \text{otherwise} \end{cases},$$

$$SOCIV_i = \begin{cases} 1 & \text{semi-skilled} \\ 0 & \text{otherwise} \end{cases}, SOCV_i = \begin{cases} 1 & \text{unskilled} \\ 0 & \text{otherwise} \end{cases}.$$

Estimating equation (B1) above by OLS resulted in a residual sum of squares (RSS) of 6.37 and an explained sum of squares (ESS) of 2.72.

- (a) Excluding the variables *SOCI*, *SOCII*, *SOCIII* and *SOCIV* from equation (1) and estimating the resultant equation by OLS, yielded an ESS of 2.48. Test the joint significance of these variables.
- (b) The Standard Industrial Classifications (SIC) defines nine single digit industry types. Explain how you would include variable(s) to control for the different industry types in equation (1). Estimating equation (B1) by OLS with the inclusion of the industry variable(s) the RSS fell to 6.08. Test the significance of this (these) variable(s).
- (c) Equation (B1) is re-estimated separately for the 278 males and the 322 females in our sample. Estimating these two models yielded RSS of 3.13 and 3.02, respectively. Comparing this model with equation (B1), carefully explain what null hypothesis you are testing, and test this hypothesis.
- (d) Adding three interacted dummy variables to equation (B1), $Female \times Educ$, $Female \times Age$ and $Female \times \ln(Income)$ the RSS fell to 6.18. Test the joint significance of these additional variables.
- (e) Compare the model estimated in (d) with the two models estimated in (c). Write out the restrictions you must impose on the model described in (c) to produce the model in (d) and test the restrictions.

Answers

- (a)
- $H_0 : \beta_4 = \beta_5 = \beta_6 = \beta_7 = 0, H_1 : \text{Any } \beta_j \neq 0$

we know that $RSS^R = TSS - ESS^R = (2.72 + 6.37) - 2.48 = 6.61$ and hence we can write:

$$F = \frac{(6.61 - 6.37)/4}{6.37/(600 - 10)} = 5.56 \quad F_{4,\infty}^{0.05} = 2.37 \quad \text{Reject } H_0 \text{ at 5\% level.}$$

- (b)
- $\ln(Happy_i) = \alpha + \beta_1 \ln(Income_i) + \beta_2 Female_i + \dots + \beta_9 Age + \sum_{j=1}^8 \gamma_j IND_{ji} + \varepsilon_i$

$$H_0 : \gamma_1 = \gamma_2 = \dots = \gamma_8 = 0, \quad H_1 : \text{Any } \gamma_j \neq 0$$

$$F = \frac{(6.37 - 6.08)/8}{6.08/(600 - 18)} = 3.47 \quad F_{8,\infty}^{0.05} = 1.94 \quad \text{Reject } H_0 \text{ at 5\% level.}$$

- (c)
- $H_0 : \beta_1^M = \beta_1^F, \beta_3^M = \beta_3^F, \dots, \beta_9^M = \beta_9^F$

$$F = \frac{(6.37 - (3.13 + 3.02))/8}{(3.13 + 3.02)/(600 - 18)} = 2.60 \quad F_{8,\infty}^{0.05} = 1.94$$

Reject H_0 at 5% level.

- (d)

$$\ln(Happy_i) = \alpha + \dots + \beta_9 Age + \gamma_1 (Fem \times Educ) + \gamma_2 (Fem \times Age) + \gamma_3 (Fem \times Income) + \varepsilon_i$$

$$H_0 : \gamma_1 = \gamma_2 = \gamma_3 = 0 \quad F_{3,\infty}^{0.05} = 2.60$$

$$F = \frac{(6.37 - 6.18)/3}{6.18/(600 - 13)} = 6.02 \quad \text{Reject } H_0 \text{ at 5\% level.}$$

- (e)
- $H_0 : \beta_3^M = \beta_3^F, \dots, \beta_7^M = \beta_7^F, F_{5,\infty}^{0.05} = 2.21$

$$F = \frac{(6.18 - 6.15)/5}{6.15/(600 - 18)} = 0.57 \quad \text{Do not reject } H_0 \text{ at 5\% level.}$$

C Exercise of Predictive failure

An insurance company is interested in the relationship between the level of insurance premiums and the value of insurance claims, particularly in light of a new premium structure they instituted at the beginning of 2021. They estimate the following model

$$c_i = \beta_0 + \beta_1 p_i + \beta_2 Q_{1i} + \beta_3 Q_{2i} + \beta_4 Q_{3i} + \varepsilon_i \quad (C1)$$

where c_i is the value of insurance claims per person insured (£ per person) at time i ; p_i is the average premium (£) at time i , $Q_{ji} = \begin{cases} 1 & \text{in quarter } j \\ 0 & \text{otherwise} \end{cases}$, $j=1,2,3$.

The model was estimated initially using quarterly data from 1993:1 to 2020:4 (prior to the introduction of the new premium structure). The following results were obtained:

$$\hat{c}_i = \underset{(0.118)}{0.752} + \underset{(0.011)}{0.918}p_i - \underset{(0.006)}{0.084}Q_{1i} - \underset{(0.006)}{0.058}Q_{2i} - \underset{(0.006)}{0.038}Q_{3i}$$

RSS = 0.057, TSS = 4.140 (Standard errors in parentheses)

Data were collected for the first eight quarters after the introduction of the new premium structure. Eight dummy variables were added to equation (C1), where $D_{01:1i}$ takes the value 1 in 2021:1 and 0 otherwise, $D_{01:2i}$ takes the value 1 in 2021:2 and 0 otherwise, etc. The new model was re-estimated using data from 1993:1 to 2022:4. The coefficients and standard errors on the dummies were as follows:

Variable	Coefficient	Standard Error
$D_{21:1i}$	0.021	0.024
$D_{21:2i}$	0.007	0.014
$D_{21:3i}$	0.011	0.013
$D_{21:4i}$	0.009	0.024
$D_{22:1i}$	0.006	0.009
$D_{22:2i}$	-0.012	0.012
$D_{22:3i}$	-0.014	0.018
$D_{22:4i}$	-0.009	0.010

- Interpret the coefficient and standard error on $D_{21:1i}$.
- What would the coefficient and standard error on p_i be in the new equation? What would the RSS of the new equation be? In both cases, explain why.
- Estimating over 1993:1 to 2022:4 without the eight dummies $D_{21:1i}$ to $D_{22:4i}$ gave an RSS of 0.065. Test whether the introduction of the new premium structure affected the relationship between the value of claims and the average premium.

Answers

- (a) $\hat{\gamma}_1$ = Prediction error for 2021:1 and $se(\hat{\gamma}_1)$ = S.E. on the prediction error.
- (b) Coefficient = 0.918 and SE = 0.011, RSS = 0.057 – as the last eight points are modelled precisely (with no error, i.e. the residual for each point would be zero) and therefore other parameters are driven by variation over the period 1993:1-2020:4 and that is why the parameter estimates are the same as that written down for the estimation of equation (C1).
- (c) $H_0 : \gamma_1 = \gamma_2 = \dots \gamma_8 = 0; H_0 : \gamma_j \neq 0$ for any j $F_{8,107}^{0.05} = 2.026$

$$F = \frac{(0.065 - 0.057)/8}{0.057/107} = 1.877 \Rightarrow \text{Do not reject } H_0$$

D Stata Commands for Structural Change

```

/* Equation 1 : Estimated over all individuals and storing output as c1*/
reg lhourpay age c.age#c.age edage if(edage>=0 & edage<=95)
est store c1
scalar rss1=e(rss)
scalar dof1=e(df_r)
/* Equation 2: Estimated over males and storing output as c3 */
reg lhourpay age c.age#c.age edage if(edage>=0 & edage<=95 & sex==1)
est store c2
scalar rss2=e(rss)
scalar dof2=e(df_r)
/* Equation 3: Estimated over females and storing output as c3 */
reg lhourpay age c.age#c.age edage if(edage>=0 & edage<=95 & sex==2)
est store c3
scalar rss3=e(rss)
scalar dof3=e(df_r)
/* Creating an excel file of the results, chow_test */
outreg2 [c1 c2 c3] using chow_test, replace excel
/* Equation 4: Estimated over all individuals including all interaction terms with gender*/
reg lhourpay age c.age#c.age edage i.sex i.sex#c.age i.sex#c.age#c.age i.sex#c.edage /*
*/ if(edage>=0 & edage<=95)
scalar rss4=e(rss)
scalar dof4=e(df_r)
/* Test the joint significance of i.sex, i.sex*age, i.sex*age^2, i.sex*edage */
testparm i.sex i.sex#c.age i.sex#c.age#c.age i.sex#c.edage
/* This save the F-test */
scalar F1a=r(F)
/* Test the joint significance of i.sex*age, i.sex*age^2, i.sex*edage */
testparm i.sex#c.age i.sex#c.age#c.age i.sex#c.edage
/* This save the F-test */
scalar F2a=r(F)
/* Equation 5: Estimated over all individuals including only additive dummy for gender*/
reg lhourpay age c.age#c.age edage i.sex if(edage>=0 & edage<=95)
scalar rss5=e(rss)
scalar dof5=e(df_r)
/* F-test comparing Equation 1 with Equation 2 + Equation 3 */
scalar F1b=(rss1-(rss2+rss3))/(dof1-(dof2+dof3))/((rss2+rss3)/(dof2+dof3))
/* F-test comparing Equation 1 with Equation 4 */
scalar F1c=(rss1-rss4)/(dof1-dof4)/(rss4/dof4)
/* F-test comparing Equation 5 with Equation 2 + Equation 3 */
scalar F2b=(rss5-(rss2+rss3))/(dof5-(dof2+dof3))/((rss2+rss3)/(dof2+dof3))
/* F-test comparing Equation 5 with Equation 4 */
scalar F2c=(rss5-rss4)/(dof5-dof4)/(rss4/dof4)
/* Reporting all 3 F-test for Chow test */

```

```
scalar list F1a F1b F1c
/* Reporting all 3 F-test for Chow test for the significance of the multiplicative terms */
scalar list F2a F2b F2c
```

E R Commands for Structural Change

#In R you must install the appropriate package. Having installed the package
 #you must load the library associated with that package each time you enter R.
 #I used the following libraries over Term 1:

```
library(tidyverse)
library(reshape2)
library(foreign)
library(ggplot2)
library(haven)
library(readxl)
library(Hmisc)
library(ggpubr)
library(rstatix)
library(groupedstats)
library(car)
library(lmtest)
library(olsrr)
library(estimatr)
library(tseries)
library(margins)
library(ivreg)
```

```
# Equation: Estimated over all individuals saving summary statistics
model15 <- lm(lhourpay ~ age + I(age^2) + edage1,
              data = subset(qlfs18, edage1>=0 & edage1<=95))
summary(model15)
res15 <- resid(model15)
rss15 <- sum(res15^2)
n15 <- nobs(model15)
k15 <- length(model15$coefficients)-1
df15 <- n15 - k15 - 1

# Equation: estimated over males and storing some values
model16 <- lm(lhourpay ~ age + I(age^2) + edage1,
              data = subset(qlfs18, edage1>=0 & edage1<=95 & sex=="Male"))
summary(model16)
res16 <- resid(model16)
rss16 <- sum(res16^2)
n16 <- nobs(model16)
k16 <- length(model16$coefficients)-1
df16 <- n16 - k16 - 1

# Equation estimated over females and storing some values
model17 <- lm(lhourpay ~ age + I(age^2) + edage1,
              data = subset(qlfs18, edage1>=0 & edage1<=95 & sex=="Female"))
```

```

summary(model17)
res17 <- resid(model17)
rss17 <- sum(res17^2)
n17 <- nobs(model17)
k17 <- length(model17$coefficients)-1
df17 <- n17 - k17 - 1
# Equation estimated over all individuals including all interaction
# terms with gender and stroing values
model18 <- lm(lhourpay ~ sex*age + sex*I(age^2) + sex*edage1,
              data = subset(qlfs18, edage1>=0 & edage1<=95))
summary(model18)
res18 <- resid(model18)
rss18 <- sum(res18^2)
n18 <- nobs(model18)
k18 <- length(model18$coefficients)-1
df18 <- n18 - k18 - 1
# Testing the significance of additive and interaction terms
test18_1 <- linearHypothesis(model18, c("sexFemale = 0", "sexFemale:age=0",
    "sexFemale:I(age^2)=0", "sexFemale:edage1"), test="F")
fstat18_1a <- test18_1$F[2]
# Testing the significance of interaction terms
test18_2 <- linearHypothesis(model18, c("sexFemale:age=0", "sexFemale:I(age^2)=0",
    "sexFemale:edage1"), test="F")
fstat18_2a <- test18_2$F[2]
# Equation estimated over all individuals including only additive dummy for gender
model19 <- lm(lhourpay ~ age + I(age^2) + edage1 + sex,
              data = subset(qlfs18, edage1>=0 & edage1<=95))
summary(model19)
res19 <- resid(model19)
rss19 <- sum(res19^2)
n19 <- nobs(model19)
k19 <- length(model19$coefficients)-1
df19 <- n19 - k19 - 1
# Constructing Chow tests using RSS
fstat18_1b <- (rss15-(rss16+rss17))/(df15-(df16+df17))/((rss16+rss17)/(df16+df17))
fstat18_1c <- (rss15-rss18)/(df15-df18)/(rss18/df18)
fstat18_2b <- (rss19-(rss16+rss17))/(df19-(df16+df17))/((rss16+rss17)/(df16+df17))
fstat18_2c <- (rss19-rss18)/(df19-df18)/(rss18/df18)
#Comparing different Chow test values
c(fstat18_1a, fstat18_1b, fstat18_1c)
c(fstat18_2a, fstat18_2b, fstat18_2c)
#Producing some nice tables of results
stargazer(model15, model16, model17, type='text', digits = 2,
          title = 'Table 2: Chow test',
          style = 'qje',

```

```
column.labels = c('All', 'Males', 'Females'), dep.var.labels = 'ln(Hourpay)')
```

References

- Dougherty, C. (2016). *Introduction to Econometrics*. OUP Catalogue. Oxford University Press.
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- Wooldridge, J. M. (2013). *Introduction to Econometrics*. Cengage.